Probability

↳ It is a measure of the likelihood that a particular event will occur (ranges from 0 to 1 to 1).

(1) Theoretical Probability
(2) Empirical Probability (experimental)

Terms:

- Sample space (S): All possible outcomes:
- Event (E): Subset of outcomes:
- P(E): Probability of event E.

$$P(E) = \underline{\text{no}} \boxed{\frac{n(E)}{n(S)}}$$

→ sample space
→ total no of outcomes

Eg #1: Roll a dice, what is probability of getting 5?

$S = \{1, 2, 3, 4, 5, 6\}$

$P(E) = \frac{1}{6}$

↳ $= \frac{1}{6}$

Types of Event

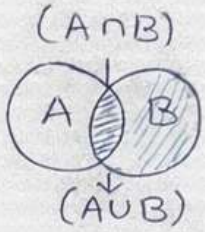
a) Certain Event (Always Happens)
b) Impossible Event (Cannot Happen)

Introductions:

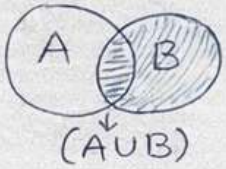
- $P(E)) = 1 \rightarrow$ Certain Event
- $P(E) = 0 \rightarrow$ Impossible Event

Instructions

- Intersection $P(A \cap B)$
  - ↳ Probability that A and B occur.



- Union $P(A \cup B)$
  - ↳ Probability that A or B occurs (either one)



Types of Event

- Mutually Exclusive (Disjoint)
  - ↳ Cannot happen at same time ?

$$P(A \cap B) = 0$$

- Independent Event: one event does affect the probability of other

$$[\, P(A \cap B) = P(A)P(B)$$

(-) Complementary Event $\rightarrow$ $\boxed{P(A) = 1 - P(A)}$

Rules:

a) $\bar{P}(\bar{A}) = 1 - P(A)$

b) Addition Rule:

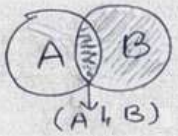
→ mutually exclusive events:

$$P(A \cup B) = P(A) + P(B)$$

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$

b) Non mutually exclusive events

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$



eg Roll a dice, what is the probability that you get an even number or a 5?

A: (2, 4, 6), E = even

B: {5}

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{3}{6} + \frac{1}{5} = \frac{2}{6}$$

$$= \frac{4}{6} = \frac{3}{3}$$

Rules:

b) Multiplication Rule

(1) Independent Event

$$\times \boxed{P(A \cap B) = P(A) \times P(B)}$$

(2) Dependent Event

$$\boxed{P(A \cap B) = P(A) \times P(B \mid A)}$$

eg a bag contains 5 balls which are 1, 2, 3, 4, 5.
Two are drawn, without replacement.
Find probability that both times an odd is drawn.

S: {(1,2), (1,2), (1,4), (1,5), (2,3),
total = (3,4), (2,5), (3,5), (3,4), (3,5), (4,5)}

S = 20

$$P(A_1 \text{ and } A_2) = P(A) \times P(A_2 \mid A_1)$$

$$= \frac{3}{5} \times \frac{2}{4} = \boxed{\frac{1}{5}}$$

i) Box contains 4 white balls & 6 black balls, one ball drawn at random & not replaced. Find probability that a black ball is found.

$$P(A \cap B) = P(A) \times P(B|A)$$

$$= \frac{6}{10} \times \frac{5}{9} = \frac{30}{9}$$

$$= \frac{30}{90} = \underline{\underline{\frac{1}{3}}}$$

ii) Two coins are tossed, probability getting at least one head.

↳ $S = \{HH, HT, TH, TT\}$

$N(E^{\sim}) = \{TT\}$

$P(E) = 1 - P(E^{\sim}) = \frac{1}{1} - \frac{1}{4} = \underline{\underline{\frac{3}{3}}}$

iii) Single card is drawn from standard deck of 52 cards.

↳ event A = face card (J, Q, K) $3 \times 4 = 12$

event B = heart

$P(A \cup B) = P(A^{\sim}) + P(B) - P(A \cap B)$

iii) Single card is drawn from standard deck of 52 cards.

↳ event A = face card (J, Q, K) $3 \times 4 = 12$

event B = heart

$$P(A \cup B) = \left[\frac{13}{52} + \frac{13}{52} - \frac{3}{52} = \boxed{\frac{22}{26}}\right.$$

Conditional Probabillity

G Probability of event B erent B occurring, given that event A has already occurred: acuracly

$$\boxed{P(B|A) = \frac{P(A \cap B)}{P(A)}} \quad \left[P(\tfrac{A}{A}) \doteq P(\tfrac{B}{M})\right]$$

Accuracy dua occurred.

Law :) Total probability

$$P(B) = P(C_1)P(C_1) \pm + P(C_2)P(B|C_2) + \cdots + P(C_n)P(\tfrac{B}{n})]$$

$$[L \rightarrow C_2, \text{---} C_n]$$

$\boxed{\text{Bayes theorem:}}$

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

where: $P(A|B)$ = probability of B given that event B
= $P(A)$ = probability of
$P(A)$ = probability of event A.
$P(B)$ = total probability of B

Eg #3 Email filtering system:

- 20% email spam → $p(\text{spam}) = 0.2$ -
- when correct match occurs, $P(E|\text{spam}) = 0.9$,
- when independent, $P(E|\text{no spam}) = 0.05$

Eg #3 Email filtering system:

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→ Find probability of actual spam:

$$P C_1 | = C_2 = P = \frac{P(\text{spam}|E)}{P(E)}$$

$$= \frac{0.9 \cdot 0.2 / 0.9 + 0.05 \cdot 0.8}{P(B) = 0.22}$$

$$P(B_2|n) = \frac{P(B_2)\,P(n_2)}{P(n)}$$

$$= \frac{0.3 \cdot 0.1}{0.78} = \frac{0.03}{\underline{\underline{38}}} = \underline{\underline{\frac{1}{26}}}$$

i) Two producing machine

↳ m_1 produces 70%, m_2 produces 30%

Defect Rate:

- m_1 – 0.4-defective
- m_2 – 0.1-defective

If a pen selected at random is defective, what is the probability that it was made by machine 2?

<u>Sol:</u>

$$P(B_2|n) = \frac{P(n|B_2) \cdot P(B_2)}{P(n)}$$

$$P(n) = P(n|B_1) \cdot P(B_1) + P(n|B_2) \cdot P(B_2)$$

$$= 0.2 \cdot 0.7 + 0.1 - 0.3$$

$$= 0.14 + 0.03$$

$$= 0.17 = \underline{\underline{7.78}}$$

$$P(B_2|n) = \frac{0.3 \cdot 0.1}{0.78} = \frac{0.09}{78} = \underline{\underline{\frac{3}{26}}} = \underline{\underline{\frac{1}{14}}}$$

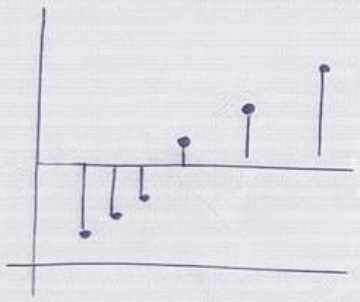
Random Variables

↳ Is a quantity (numerical) that depends on the outcome of a random experiment

- Discrete RV: countable values (no fractions)
- Continuous RV: any value in Range (fractions)
- weighted mean (some values contribute more than other stuff)

Variance
Standard Deviation

} let measure how spread out the values are from the mean:



$$var(X) = \frac{\sum (X_i - \mu)^2}{n}$$

$$\text{std dev} = \sqrt{\frac{\sum (X_i - \mu)^2}{n}}$$

Probability Distributions

↳ It describes how probabilities are assigned to all possible outcomes.

- how likely different outcomes are

Types

↳ Discrete PD

eg Binomial Distribution

↳ Continuous PD

- eg Uniform Distribution
- Normal Distribution

Binomial Distribution

↳ Used to model the no. or of successes in n trials.

l'probability of getting an event scucess exactly once.

$$\boxed{P(X = k) = p^k (1-p)^{n-k}}$$

where $n = \frac{n.n!}{k!} = \frac{n\,k}{k!}(n-k)!)$

p = probability of success.

nC_k = k ways of choocing k successes out n items.

Rules to be followed

i) There are a fixed no. of trials:

ii) Each trial has only 2 outcomes:

iii) Probability of success is same every time;

iv) Trials are independent.

Application of BD

↳ Spam detection

↳ Cancer treatment world

Eg A fair coin is tossed 5 times.

Find probability that exactly 3 Heads are obtained:

given $P(3=3) = \frac{5^{C_2}}{3\underline{3}}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^{5-3}$

$n=5$

$k=3$

$P(\text{Head} = \frac{1}{2}$

$= \frac{5 \cdot 4 \cdot 3}{3 \cdot 2 \cdot 1} = \frac{10}{10}$

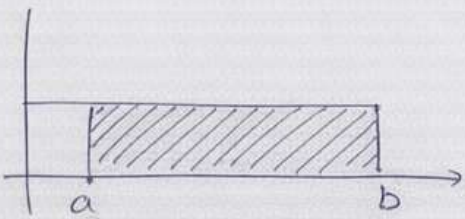
$= \frac{10}{10}\left(\frac{1}{2}\right)^2$

$\frac{5}{16}$

Uniform Distribution

↳ it is a PD where all outcomes are "equally likely".

- $\text{Prob}(X = x) = \frac{1}{b-a} \quad \{ a \leq x \leq b$



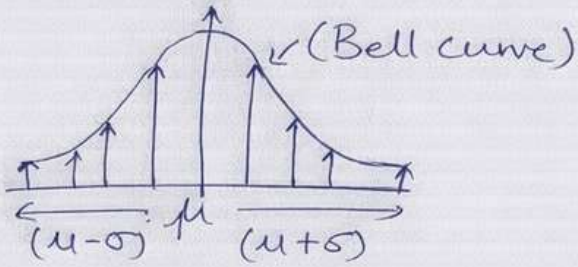
(probability)

$\text{PDF} = f(x) = \frac{1}{b-a}$

$\text{Prob}(a \leq x \leq b) = \frac{1}{b-a}$

Normal Distribution (Gausian Distribution)

↳ It is a continuous PD that is bell-shaped and symmetric.



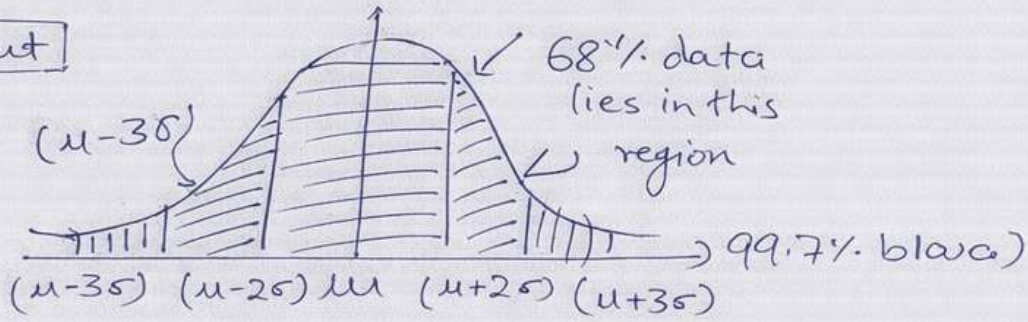
Characteristics

i) mean = median = mode (curve center)

ii) Total area under curve = 1

iii) μ: mean, σ = std dev (Spreed the curve)

Shortcut



PDF

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{(x-\mu)^2}{2\sigma^2}}$$

μ = mean

σ = std dev

$e = 2.71$

→ Key point of central limit

i) works for any population mean, sample pop –

ii) population mean (μ), $(\bar{x})$

i) $\sigma = \frac{\sigma}{\sqrt{n}}$

Central Limit Theorem (more data it tends to normal)

↳ As we draw more data it tends to normal.

·) Take a large no. of random samples form any dataset (finite & complex distribution). Large sample. Large sample suppots required distribution approx. by normal.